

Plugging into the Future: An Exploration of Electricity Consumption Pattern

Problem Definition

Electricity plays a vital role in India's economic and social development. The COVID-19 pandemic caused significant changes in electricity demand due to lockdowns and changes in industrial and residential activities. This project aims to analyze electricity consumption across Indian states and regions during 2019–2020 to understand consumption trends, regional differences, and the impact of the pandemic.

Problem Understanding

The objective of this project is to study electricity consumption patterns using Tableau. The analysis helps identify high and low electricity-consuming states, compare regional consumption, and evaluate changes before and during the COVID-19 pandemic. Interactive dashboards and visualizations make the analysis easy to understand.

Data Collection

The dataset was collected from a publicly available source in CSV format. It contains electricity consumption data for Indian states from January 2019 to December 2020.

Dataset Fields

States

Regions

Dates

Latitude

Longitude

Usage (Electricity Consumption)

Data Extraction and SQL Operations

The dataset was imported into the database and stored in a table named Consumption. SQL queries were performed to retrieve records, calculate total electricity consumption, group data by states and regions, and verify the accuracy of the dataset before visualization.

Connecting Database with Tableau

The cleaned dataset was connected to Tableau Public. Geographic roles were assigned to the States field, dates were converted into date format, and measures were validated before creating visualizations.

Data Preparation

The dataset was prepared by:

Removing missing values.

Correcting state names that Tableau could not recognize.

Assigning Geographic Role as State/Province.

Formatting the Dates column.

Verifying latitude and longitude values.

Filtering data into 2019 and 2020 where required.

Data Visualizations

Visualization 1: 2019 Electricity Consumption

Chart Type: Filled Map

This visualization displays the electricity consumption of each Indian state during 2019. The map provides a geographical overview of electricity demand before the COVID-19 pandemic. States with darker colors represent higher electricity consumption, while lighter colors indicate lower consumption. This visualization serves as the baseline for comparing changes in 2020.

Insight Industrialized and highly populated states consumed more electricity, whereas smaller states and Union Territories recorded comparatively lower usage.

Visualization 2: 2020 State Consumption

Chart Type: Filled Map

This map represents electricity consumption across Indian states during 2020, the period affected by the COVID-19 pandemic. It helps identify how electricity demand changed due to lockdowns and reduced industrial activity.

Insight Several states experienced reduced electricity consumption during lockdown months, while some regions showed stable demand due to residential electricity usage.

Visualization 3: Total Consumption

Chart Type: Bar Chart

This visualization compares the total electricity consumption of all states during the study period. States are arranged in descending order, making it easy to identify the highest and lowest electricity consumers.

Insight The chart clearly highlights major electricity-consuming states and helps compare electricity demand among all Indian states.

Visualization 4: Usage by Region

Chart Type: Pie Chart

This visualization illustrates the percentage contribution of each region (Northern, Southern, Eastern, Western, and North Eastern) to the total electricity consumption of India. The pie chart provides an overall understanding of regional electricity distribution.

Insight The Southern and Western regions contribute a larger share of electricity consumption because of their higher industrial and urban development.

Visualization 5: Total Region Consumption

Chart Type: Horizontal Bar Chart

This visualization compares the total electricity consumption across all regions of India. It allows users to easily identify which region consumes the highest and lowest amount of electricity.

Insight Regional consumption differences reflect variations in industrial activities, climate conditions, infrastructure development, and population density.

Dashboard

An interactive Tableau dashboard was created by combining all five visualizations. Users can analyze electricity consumption using filters such as Year, Region, and State. The dashboard provides a complete overview of electricity consumption patterns across India.

Dashboard Features

Interactive filters

State-wise comparison

Region-wise comparison

Geographic visualization

User-friendly design

Stories

Story 1 – 2019 Electricity Consumption

Shows state-wise electricity consumption across India during 2019, providing a pre-pandemic baseline for analysis.

Story 2 – 2020 State Consumption

Displays electricity consumption during 2020, highlighting the impact of COVID-19 on different states.

Story 3 – Total Consumption

Compares total electricity consumption across all states to identify the highest and lowest consumers.

Story 4 – Usage by Region

Illustrates each region's percentage contribution to India's total electricity consumption.

Story 5 – Total Region Consumption

Compares electricity consumption across regions, revealing differences in regional energy demand.

Performance Testing

The Tableau dashboard was tested for performance by ensuring that all visualizations loaded correctly, filters responded quickly, and the dataset rendered efficiently. The dashboard remained interactive while switching between different years and regions.

Key Findings

Electricity consumption varies significantly across Indian states.

Maharashtra, Gujarat, Tamil Nadu, and Uttar Pradesh are among the highest electricity-consuming states.

Western and Southern regions contribute the largest share of electricity consumption.

Electricity demand declined during the COVID-19 lockdown and gradually recovered during the later months of 2020.

Geographic visualization helps identify regional differences in electricity usage.

Conclusion

This project successfully analyzed electricity consumption patterns across India during 2019 and 2020 using Tableau. The dashboard and story effectively demonstrate state-wise and region-wise electricity usage, highlight the impact of the COVID-19 pandemic, and provide meaningful insights into energy consumption trends. The findings can support policymakers, researchers, and electricity providers in understanding demand patterns and planning future energy management strategies.